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Abstract: Abstract...

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Process	Decay	Lower Mass Limit	Reference
$pp \to W'/Z'$	$V' \to WH/WZ/ff$	5.5 TeV	[1]
$pp \to Z'$	$Z' \to \ell^+\ell^-$	5.2 TeV	[2]
$pp \to W'$	$W' \to \ell\bar{\nu}_\ell$	5.7 TeV	[3]

Table 1. Current most stringent experimental limits on Z' and W' from LHC.

Contents

1 Constraints from LHC

In this section, we summarize the current LHC limits on New Physics particles. Unless specified otherwise, these results are based on data at a center-of-mass energy of 13 TeV, corresponding to an integrated luminosity of 136-140 fb^{-1}.

1.1 Z' and W'

In the early 2026, CMS has published their combined statistical analysis of searches for heavy vector boson resonances [1]. Both Drell–Yan (DY) and Vector Boson Fusion (VBF) data are analyzed. For Z', the decay channels $\ell^+\ell^-$, $q\bar{q}$, $t\bar{t}$, ZZ, Zh are considered. For W', the decay channels $\ell\bar{\nu}_\ell$ ($\ell = e, \mu, \tau$), $q\bar{q}'$, $t\bar{b}$, WZ, Wh are considered. This combined analysis gives the most stringent constraints of Heavy Vector Triplet (HVT), where Z' and W' are excluded with masses below 5.5 TeV.

For sequential standard model, the most stringent limits on Z' bosons come from high-mass dilepton resonance searches, which set a lower mass limit on 5.2 TeV [2], while the W' has a lower limit on 5.7 TeV, coming from $\ell\bar{\nu}_\ell$ channel [3].

We summarize these constraints in Table 1.

Process	Decay	Mass Range	Reference
$pp \to h$	$h \to AA \to 4\gamma$	$0.1 - 1.5$ GeV	[5]
$pp \to A$	$A \to \tau\tau$	$20 - 90$ GeV	[7]
$pp \to h$	$h \to AA \to \mu\mu\tau\tau$	$3.6 - 21$ GeV	[8]

Table 2. Current most stringent experimental limits on light A from LHC.

1.2 CP-odd Neutral Higgs Boson (A) and CP-even Neutral Higgs Boson (H)

LHC constraints on CP-odd neutral Higgs boson are evolved into two main regimes: light pseudoscalars (masses below the 125 GeV Higgs boson) and searches for heavy pseudoscalars (masses up to several TeV).

For particle A mass $m_A \leq 1$ GeV, the decay mode $A \to \gamma\gamma$ become important because it involves in photophilic scenarios [4]. CMS [5, 6] has studied the $pp \to h \to AA \to 4\gamma$ process. The result set a upper limit on the branching ratio $\mathcal{B}(h \to AA \to 4\gamma)$ down to $\mathcal{O}(10^{-3})$. When $m_A \geq 2m_\tau$, the decay $A \to \tau^+\tau^-$ becomes dominant. Ref. [7] reports a search for $pp \to A \to \tau^+\tau^-$ process. One τ decays to electron and neutrinos, while the other τ decays to muon and neutrinos. A mass range between 20 GeV and 90 GeV has been searched, and no significant excess is observed. Recently, Ref. [8] has searched for $h \to AA \to \mu^+\mu^-\tau^+\tau^-$ decay. Four decay modes $\tau\tau \to e\mu +$ neutrinos, $\tau\tau \to e +$ hadron(s) $+$ neutrinos, $\tau\tau \to \mu +$ hadron(s) $+$ neutrinos, and $\tau\tau \to$ hadrons $+$ neutrinos are considered. For $m_A \in (3.6, 21)$ GeV, the branching ratio is set to upper limit on $\mathcal{O}(10^{-4})$. Table 2 summarizes the searches for light A.

For the heavy A in TeV level, the Two-Higgs-Doublet Model (2HDM) is the most popular benchmark model in searching for A, as well as the CP-even Higgs boson H. In the 2HDM, the ratio of the vacuum expectation values (VEV) of two doublets, denoted as $\tan\beta$, is an important parameter. When $\tan\beta$ is large, say $10 \leq \tan\beta \leq 60$, the dominant decay modes of A/H are $A/H \to b\bar{b}$ or $\tau^+\tau^-$. Early searches [9, 10] for $pp \to A/H \to \tau^+\tau^-$ give strong constraints on m_A and m_H. The lower mass limits on m_A and m_H are set on around 2 TeV when $\tan\beta \geq 50$. For the low $\tan\beta$, the decay mode $A/H \to t\bar{t}$ becomes important since the coupling to top quarks is enhanced. Ref. [11] has searched for the $pp \to A/H \to t\bar{t}$ process, where one top quark decays leptonically and the other hadronically. Assuming $m_A = m_H$, the masses up to 1240 GeV are excluded for the lowest tested $\tan\beta = 0.4$. When $m_A > m_H + m_Z$, the decay $A \to ZH$ becomes possible and is dominant in a wide range of 2HDM parameter space. Ref. [12] has searched for $pp \to A \to ZH$ process, with $H \to t\bar{t}$. Z boson further decays to muons or electrons, while top quarks decay hadronically. The upper limits on $m_A = 2100$ GeV and $m_H = 2000$ GeV are obtained. The summary of constraints on heavy A and H are listed in table 3.

1.3 $H^\pm$

The searches for $H^\pm$ are categorized by its mass relative to the top quark: light ($m_{H^\pm} < m_t$) and heavy ($m_{H^\pm} > m_t$).

For the light $H^\pm$, top quark can decay to $H^\pm$ and b quark, and $H^\pm$ further decays to Standard Model particles. ATLAS has reported three decay modes: $H^+ \to \tau^+\nu_\tau$ [13],

Process	Decay	Lower Mass Limit	Reference
$pp \to A/H$	$A/H \to \tau^+\tau^-$	2000 GeV	[9, 10]
$pp \to A/H$	$A/H \to t\bar{t}$	1240 GeV	[11]
$pp \to A \to ZH$	$H \to t\bar{t}, Z \to \ell^+\ell^-$	$m_A = 2100$ GeV, $m_H = 2000$ GeV	[12]

Table 3. Current most stringent experimental limits on light A from LHC.

Process	Decay	Mass Range	Reference
$t \to H^\pm b$	$H^+ \to \tau^+\nu_\tau$	$80 - 160$ GeV	[13]
$t \to H^\pm b$	$H^\pm \to cs$	$60 - 168$ GeV	[14]
$t \to H^\pm b$	$H^\pm \to cb$	$60 - 160$ GeV	[15]
$pp \to tbH^\pm$	$H^\pm \to tb$	200-2000 GeV	[16]
$pp \to tbH^\pm$	$H^\pm \to Wh$	230-3000 GeV	[17]
$pp \to WbH^\pm b$	$H^\pm \to \tau\nu_\tau$	80-3000 GeV	[13]

Table 4. Current most stringent experimental limits on $H^\pm$ from LHC.

Process	Decay	Lower Mass Limit	Reference
$pp \to H^{++}H^{--}$	$H^{\pm\pm} \to \ell^\pm\ell^\pm$	1.08 TeV	[18]
$pp \to H^{\pm\pm}jj$	$H^{\pm\pm} \to W^\pm W^\pm$	1.5 TeV	[19]

Table 5. Current most stringent experimental limits on $H^{\pm\pm}$ from LHC.

$H^\pm \to cs$ [14] and $H^\pm \to cb$ [15]. For $H^\pm \to \tau\nu_\tau$, current limits on the branching fraction $\mathcal{B}(t \to H^\pm b) \times \mathcal{B}(H^\pm \to \tau\nu_\tau)$ are as low as 0.02% to 0.27% with $m_{H^\pm}$ from 80 GeV to 160 GeV. For $H^\pm \to cs$, upper limits on $\mathcal{B}(t \to H^\pm b)$ in the cs channel between 0.07% and 3.6%, assuming $\mathcal{B}(H^\pm \to cs) = 1$ and $m_{H^\pm} \in (60, 168)$ GeV. Similar to $H^\pm \to cs$, $H^\pm \to cb$ gives the upper limits on $\mathcal{B}(t \to H^\pm b)$ between 0.15% and 0.42%, where $m_{H^\pm} \in (60, 160)$ GeV.

For the heavy $H^\pm$, the $pp \to tbH^\pm$ has been widely studied since it is the dominant process of 2HDM model. The decay channels $H^\pm \to tb$ [16], $H^\pm \to Wh$ [17], and $H^\pm \to \tau\nu_\tau$ [13]. The mass range from 200 GeV to 3 TeV has been searched. The constraints for $H^\pm$ are summarized in Table. 4.

1.4 $H^{\pm\pm}$

At the LHC, doubly charged Higgs bosons can be produced via DY process or VBF process. Then they can decay to same signed leptons or vector bosons. Ref. [18] reports the search for DY production and the leptonic decay of $H^{\pm\pm}$. All combinations of lepton pairs have been considered, and they have the same branching ratio (1/6). This analysis gives a lower mass limit on 1.08 TeV. In another analysis [19], the VBF production and $H^{\pm\pm} \to W^\pm W^\pm$, where W further decays to lepton and neutrino, is studied. In this case, the mass range from 0.2 TeV to 1.5 TeV is excluded. The searches for doubly charged Higgs bosons are listed in table. 5.

Process	Decay	Lower Mass Limit	Reference
$pp \to e'^+e'^-, e'N, NN$	$e' \to Z(H)e, N \to W\ell$	1220 GeV	[22]
$pp \to \mu'^+\mu'^-, \mu'N, NN$	$\mu' \to Z(H)\mu, N \to W\ell$	1270 GeV	[22]
$pp \to \tau'^+\tau'^-$	$\tau' \to \tau a_\tau \to \tau\gamma\gamma$	700 GeV	[23]
$pp \to L^+L^-, LN, NN$	$L \to b\bar{b}\tau, t\bar{b}\bar{\nu}_\tau,\ N \to t\bar{t}\nu_\tau, t\bar{b}\tau^-$	910 GeV	[24]

Table 6. Current experimental limits on VLL from LHC.

1.5 Vector-Like Lepton (VLL)

VLLs are predicted in many extensions of the Standard Model, including composite Higgs models and theories addressing the muon $g-2$ anomaly. The production and decay modes for VLLs depend on the assumed $SU(2)$ representation [20, 21]. In the doublet scenario, two mass-degenerate VLLs at tree level, one electrically charged ($L^\pm$) and one electrically neutral (N), form an SU(2) doublet (L, N). At the LHC, VLLs are predominantly produced in pairs via the electroweak interaction ($pp \to Z^*/\gamma^* \to L^+L^-$, $pp \to Z^* \to N\bar{N}$ or $pp \to W^* \to L^+\bar{N}$). The charged VLL decay modes are $L \to Z\ell$ and $L \to h\ell$. When $m_L \gg m_h$, the branching ratio reaches 50% for each modes. The neutral VLL N decays to lepton and W boson with 100% branching ratio. In the singlet scenario, only the charged VLL is present. It is also produced in pairs, $pp \to Z^*/\gamma^* \to L^+L^-$. Its decay modes are $L \to W\nu$, $L \to Z\ell$ and $L \to h\ell$, with branching ratio 50%, 25% and 25%, respectively. Ref. [22] has searched for these processes and decay modes. Its result gives lower mass limits on 1220 GeV (1270 GeV) and 320 GeV (400 GeV) for vector-like electrons (muons) in the doublet and singlet scenarios, respectively.

In another paper [23], the pair production of vector-like tau (τ') has been considered. τ' decays to SM τ and a light long-lived pseudoscalar boson (a_τ). Since a_τ is long-lived particle, it leads to deposits in the muon system, which motivated experimentalists to search for this process. In the muon system, a_τ further decays to two photons, assuming it has mass below the $\tau^+\tau^-$ threshold. In this analysis, the mass of a_τ is assumed to be $m_{a_\tau} = 2$ GeV. This analysis targets signals with at least one hadronically decaying τ and with at least one muon detector shower resulting from the a_τ decay in the CMS muon system. Its result excludes τ' masses up to 700 GeV.

Ref. [24] searches for VLL in the 4321 model. In this model, only one generation of VLLs is present in the model, and it is assumed to be fully mass degenerate. VLLs are produced through the pair production as mentioned above, and then they decay through a vector leptoquark U_1 as $L \to U_1\bar{b}$ and $N \to \bar{U}_1 t$, followed by $U_1 \to t\bar{\nu}_\tau$ or $U_1 \to b\tau^+$. The branching ratios are $\mathcal{B}(L \to b\bar{b}\tau^+) = \mathcal{B}(L \to t\bar{b}\bar{\nu}_\tau) = 50\%$ and $\mathcal{B}(N \to t\bar{t}\nu_\tau) = \mathcal{B}(N \to t\bar{b}\tau^-) = 50\%$. Ref. [24] set a lower observed (expected) limit of 910 GeV on the VLL mass.

Table 7 summarizes the current LHC constraints on VLL.

1.6 Long-Lived Particles

Long-lived particles (LLP) appear in many dark matter and SUSY scenarios. These particles can travel a measurable distance through a particle detector before decaying. Detecting LLPs is a challenge because their decay signatures are non-conventional. Usually some specific reconstruction algorithms are necessary.

Ref. [25] presents a search for LLPs by ATLAS detector. This analysis evaluates three types of LLPs: gluinos ($\tilde{g}$), charginos ($\tilde{\chi}_1^{\pm}$), and staus ($\tilde{\tau}$). All of these particles are assumed to be pair-produced. In these process, each gluino decays into a stable neutralino ($\tilde{\chi}_1^0$) and two quarks, each chigino decays into a stable neutralino and a pion, and each stau decays into a τ lepton and a stable gravitino ($\tilde{G}$). Since no new physics is observed, 95% Confident Level (CL) mass limits are: $m_{\tilde{g}} > 2270$ GeV with a lifetime $\tau \approx 30$ ns, $m_{\tilde{\chi}_1^{\pm}} > 1300$ GeV ($\tau > 100$ ns), and $m_{\tilde{\tau}} > 560$ GeV ($\tau = 10$ ns).

Ref. [26] extends previous study to events with one jet. They reconstruct the signal that recoils a high-momentum jet from initial state radiation. In this case, chargino decays into pion and neutralino, $\tilde{\chi}_1^{\pm} \to \pi^{\pm}\tilde{\chi}_1^0$. When the lifetime of chargino is shorter than 0.03 ns, the mass up to 225 GeV is excluded. When the lifetime is as long as 1 ns, the mass up to 720 GeV is excluded for the Higgsino scenario and 880 GeV is excluded for the wino scenario. For the stau, two scenarios are considered: the Constrained Minimal Supersymmetric Standard Model (CMSSM) and the gauge-mediated supersymmetry-breaking (GMSB) model. The CMSSM $\tilde{\tau}$ decays into $\tau\tilde{\chi}_1^0$ with 100% branching fractions, while the GMSB $\tilde{\tau}$ has 100% branching fraction for $\tilde{\tau} \to \tau\tilde{G}$. The mass of stau up to 320 GeV is excluded for the CMSSM scenario and 300 GeV is excluded for the GMSB scenario.

Ref. [27] presents a search for LLPs by targeting at least one displaced vertex in event. In this paper, a new dataset with $\sqrt{s} = 13.6$ TeV and $\mathcal{L} = 164$ fb^{-1} is used. Three types of LLPs are searched: Higgsino with lepton-number-violating (L-violating) decay, Higgsino with baryon-number-violating decay (B-violating), and stop ($\tilde{t}$). Both of them are pair productions. For the Higgsino, the following production modes are considered: $\tilde{\chi}_1^0\tilde{\chi}_2^0$, $\tilde{\chi}_1^0\tilde{\chi}_1^{\pm}$, $\tilde{\chi}_2^0\tilde{\chi}_1^{\pm}$, and $\tilde{\chi}_1^{\pm}\tilde{\chi}_1^{\mp}$, where $\tilde{\chi}_1^0$ is the lightest supersymmetric particle, $\tilde{\chi}_2^0$ is the next-to-lightest neutralino and $\tilde{\chi}_1^{\pm}$ is the lightest chargino. When the Higgsino is L-violating type, the neutral LLPs decay into $\mu^- u\bar{d}$, and $\tilde{\chi}_1^{\pm} \to \mu^{\pm} d\bar{d}$. When the Higgsino is B-violating type, the neutral LLPs decay into tsb, and $\tilde{\chi}_1^{+} \to \bar{b}\bar{s}\bar{b}$. For stop, it decays into $\mu^+ b$. Results of this paper set a 95% CL lower limits on these particles: Higgsino with L-violating decays: 1600 GeV, Higgsino with B-violating decays: 1150 GeV, stop: 1850 GeV. The lifetimes of these particles are assumed to be 0.1 ns.

A Some title

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Acknowledgments

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Process	Decay	Lower Mass Limit	Reference
$pp \to \tilde{g}\tilde{g}$	$\tilde{g} \to \tilde{\chi}_1^0 qq$	2270 GeV (30 ns)	[25]
$pp \to \tilde{\chi}_1^\pm \tilde{\chi}_1^\mp$	$\tilde{\chi}_1^\pm \to \tilde{\chi}_1^0 \pi$	1300 GeV (100 ns)	[25]
$pp \to \tilde{\tau}^+\tilde{\tau}^-$	$\tilde{\tau} \to \tau \tilde{G}$	560 GeV (10 ns)	[25]
$pp \to \tilde{\chi}_1^0 \tilde{\chi}_1^\pm j$	$\tilde{\chi}_1^\pm \to \tilde{\chi}_1^0 \pi$	225 GeV (0.03 ns)	[26]
$pp \to \tilde{\chi}_1^0 \tilde{\chi}_1^\pm j$	$\tilde{\chi}_1^\pm \to \tilde{\chi}_1^0 \pi$	720 GeV (1 ns)	[26]
$pp \to \tilde{\chi}_1^0 \tilde{\chi}_1^\pm j$	$\tilde{\chi}_1^\pm \to \tilde{\chi}_1^0 \pi$	880 GeV (1 ns, wino)	[26]
$pp \to \tilde{\tau}^+\tilde{\tau}^-$	$\tilde{\tau} \to \tau \tilde{\chi}_1^0$	320 GeV (1 ns)	[26]
$pp \to \tilde{\tau}^+\tilde{\tau}^-$	$\tilde{\tau} \to \tau \tilde{G}$	300 GeV (1 ns)	[26]
$pp \to \tilde{\chi}_1^0\tilde{\chi}_2^0, \tilde{\chi}_1^0\tilde{\chi}_1^\pm, \tilde{\chi}_2^0\tilde{\chi}_1^\pm, \tilde{\chi}_1^\pm\tilde{\chi}_1^\mp$	$\tilde{\chi}^0 \to \mu^- u\bar{d}, \tilde{\chi}_1^\pm \to \mu^\pm d\bar{d}$	1600 GeV (0.1 ns)	[27]
$pp \to \tilde{\chi}_1^0\tilde{\chi}_2^0, \tilde{\chi}_1^0\tilde{\chi}_1^\pm, \tilde{\chi}_2^0\tilde{\chi}_1^\pm, \tilde{\chi}_1^\pm\tilde{\chi}_1^\mp$	$\tilde{\chi}^0 \to tsb, \tilde{\chi}_1^+ \to \bar{b}\bar{s}\bar{b}$	1150 GeV (0.1 ns)	[27]
$pp \to \tilde{t}\bar{\tilde{t}}$	$\tilde{t} \to \mu^+ b$	1850 GeV (0.1 ns)	[27]

Table 7. Current experimental limits on LLPs from LHC.

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